



OBJECT ORIENTED SOFTWARE ENGINEERING

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Validation Testing

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- The process of evaluating software **during the development process** or **at the end of the development process** to determine whether it **satisfies specified** business requirements.
- **Validation Testing** ensures that the product **actually meets the client's needs**

- Validation testing **follows integration testing**
- Validation testing is the practice of ensuring that **software meets the quality standards set by the customer** and that the **product meets customer requirements**
- Designed to ensure that
 - All functional requirements are satisfied
 - All behavioral characteristics are achieved
 - All performance requirements are attained
 - Documentation is correct
 - Usability and other requirements are met (e.g., transportability, compatibility, error recovery, maintainability).

- Validation testing is sometimes called **acceptance testing** because it is usually performed once the developer has finished writing code.
- In this sense, it is a process of **checking to see if a product can be accepted** based on certain criteria or requirements that are defined in advance by the consumer or user

- After each validation test
 - A **deviation from specification is uncovered** and a **deficiency list** is created
- Validation is also part of a larger suite of tests known as **non-functional testing**, which also includes **performance** and **usability testing**.
- Validation testing usually **occurs during the final stages of development** and **prior to releasing a new product or service to the customers**.

- Validation testing focuses on assessing whether the **software meets the intended requirements and functions correctly within its intended environment.**
- Here are some key points and aspects related to validation testing in software testing:
- **Purpose:** The main purpose of validation testing is to determine if the software product meets the intended purpose and functionality as outlined in the **requirements specification (SRS)**. It aims to validate that the software addresses the business or user needs effectively.

- **User-Centric Focus:** Validation testing is centered around the end-users or customers. It is about confirming that the software provides value to users and aligns with their expectations.
- **Requirements Validation:** During validation testing, the **software's functionality is evaluated** against the **documented requirements**. This ensures that all specified features and capabilities are present and working as intended.

- **Environment Compatibility:** Validation testing ensures that the software **performs well within the target environment**, including different hardware, operating systems, network configurations, and other relevant factors.
- **User Acceptance Testing (UAT):** User Acceptance Testing is a crucial part of validation testing. It involves end-users testing the software to confirm that it meets their needs and that they can use it effectively to achieve their tasks.

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- **Defect Reporting and Resolution:** Any defects or issues identified during validation testing are **documented** and **reported back to the development team**. The team then addresses these issues and provides fixes or enhancements as necessary.
- **Completion Criteria:** The criteria for successful completion of validation testing are often **outlined in advance**. These criteria might include specific functionality being tested, **performance benchmarks being met**, and **a minimum number of defects being identified and resolved**.

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- **Sign-Off:** Validation testing **concludes with stakeholders**, such as **product managers, business analysts, and end-users**, reviewing the testing results.
 - Once they are satisfied with the software's performance and alignment with requirements, they **provide their approval or sign-off**.



THANK YOU

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